

Production-Grade Wazuh SOC Agent Runbook

Auto-Recovering Endpoint Monitoring for Windows, Linux, macOS, Syslog, and Winlogbeat

Prepared for: SOC / SIEM Operations Team

Scope: Endpoint agent reliability, log onboarding, service recovery, and SOC validation

Document status: Production implementation template

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Control Area	Production Standard
Agent connectivity	TCP 1514, unique enrollment keys, stable DNS/IP, keepalive tuning, OS service recovery
Windows telemetry	Wazuh Agent + Security/System/Application + PowerShell + Sysmon + Defender + RDP + WMI + WinRM
Linux telemetry	Wazuh Agent + syslog/auth/secure/messages + auditd + cron + package + web/db logs
macOS telemetry	Wazuh Agent + system/install/wifi/audit + Unified Logging where required
Network devices	Syslog to Wazuh manager using TCP 514 where possible
Fallback/parallel pipeline	Winlogbeat to Logstash/Elastic/SIEM when required by architecture
SOC readiness	FIM, SCA, syscollector, validation tests, disconnected-agent alerting and daily health checks

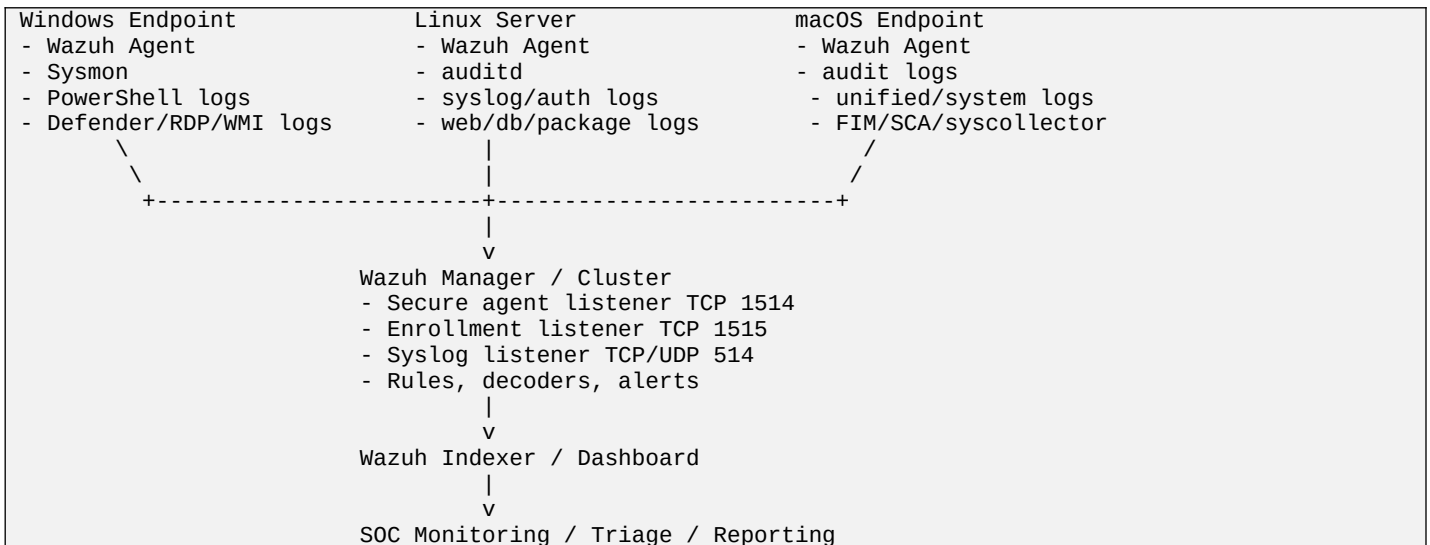
1. Executive Summary

This runbook provides a production-ready configuration standard for Wazuh agents on Windows, Linux, and macOS endpoints. It also covers syslog ingestion for network/security devices and Winlogbeat as an optional Windows event forwarding component where a Logstash/Elastic pipeline is required.

The target operational state is: whenever an endpoint and its network path are available, the monitoring service starts automatically, reconnects automatically, forwards SOC-relevant logs, and can be validated from both endpoint and Wazuh manager side.

No endpoint monitoring solution can guarantee that an agent will never disconnect under all conditions. Shutdown, sleep state, firewall block, route failure, DNS failure, duplicate agent identity, manager outage, certificate/key issues, or severe endpoint failure can still cause disconnection. This document therefore implements a self-healing and SOC-observable design.

2. Reference Architecture



3. Production Assumptions and Variables

Variable	Example	Replace With
WAZUH_MANAGER_FQDN	wazuh-manager.company.local	Your manager DNS name or load balancer FQDN
WAZUH_MANAGER_IP	10.10.10.50	Your manager IP address
AGENT_GROUP	windows-workstations / linux-servers	Your Wazuh group naming standard
SYSLOG_ALLOWED_CIDR	10.0.0.0/8	Approved device/server source ranges
LOGSTASH_IP	10.10.10.60	Only required if using Winlogbeat

4. Required Network and Firewall Rules

Source	Destination	Port	Protocol	Purpose	Action
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Wazuh Agent	Wazuh Manager	1514	TCP	Secure agent event forwarding	Allow
Wazuh Agent	Wazuh Manager	1515	TCP	Agent enrollment/auth entication	Allow during enrollment; keep if auto-enrollment is used
Network devices / syslog forwarders	Wazuh Manager	514	TCP preferred / UDP optional	Syslog ingestion	Allow from approved CIDRs only
SOC admins	Wazuh Dashboard	443 or 5601	TCP	Dashboard access	Allow via admin networks/VPN only
SOC/API users	Wazuh API	55000	TCP	API-based management/health checks	Restrict strongly

5. Wazuh Manager Production Configuration

File path on Wazuh manager: `/var/ossec/etc/ossec.conf`

Manager `ossec.conf` baseline

```
<ossec_config>
  <global>
    <jsonout_output>yes</jsonout_output>
    <alerts_log>yes</alerts_log>
    <logall>no</logall>
    <logall_json>no</logall_json>
    <email_notification>no</email_notification>

    <!-- Recommended for production endpoints. Keep lower for servers where fast outage detection
    matters. -->
    <agents_disconnection_time>30m</agents_disconnection_time>
    <agents_disconnection_alert_time>0</agents_disconnection_alert_time>
  </global>

  <!-- Wazuh agent secure communication. TCP is preferred for reliability. -->
  <remote>
    <connection>secure</connection>
    <port>1514</port>
    <protocol>tcp</protocol>
    <queue_size>131072</queue_size>
  </remote>

  <!-- Syslog listener for network/security devices. Restrict source ranges. -->
  <remote>
    <connection>syslog</connection>
    <port>514</port>
    <protocol>tcp</protocol>
    <allowed-ips>10.0.0.0/8</allowed-ips>
    <allowed-ips>172.16.0.0/12</allowed-ips>
    <allowed-ips>192.168.0.0/16</allowed-ips>
    <local_ip>0.0.0.0</local_ip>
  </remote>
</ossec_config>
```

Manager validation commands

```
sudo systemctl restart wazuh-manager
sudo systemctl status wazuh-manager --no-pager
sudo ss -tulnp | egrep '1514|1515|514'
sudo tail -n 100 /var/ossec/logs/ossec.log
```

6. Universal Wazuh Agent Connectivity Block

Apply this block to Windows, Linux, and macOS agents. Use a stable FQDN or load balancer address. If DNS is unreliable, use a static manager IP address.

Universal agent client block

```
<client>
  <server>
    <address>WAZUH_MANAGER_FQDN</address>
    <port>1514</port>
    <protocol>tcp</protocol>
    <max_retries>5</max_retries>
    <retry_interval>10</retry_interval>
  </server>
  <notify_time>30</notify_time>
  <time-reconnect>120</time-reconnect>
  <auto_restart>yes</auto_restart>
</client>
```

Setting	Production Purpose	Recommended Value
protocol	Reliable agent-to-manager communication	tcp
notify_time	Keepalive/check-in interval. Must be lower than manager disconnection time.	30 seconds
time-reconnect	Retry wait time after failed check-in. Should be higher than notify_time.	120 seconds
auto_restart	Agent restarts when valid centralized config is received.	yes
max_retries / retry_interval	Connection retry behavior for manager server entries.	5 / 10 seconds

7. Windows Production Deployment

7.1 Windows Paths

Item	Path / Service
Wazuh config	C:\Program Files (x86)\ossec-agent\ossec.conf
Wazuh log	C:\Program Files (x86)\ossec-agent\ossec.log
Wazuh service	WazuhSvc or wazuh depending package/version display
Windows EVTX directory	C:\Windows\System32\winevt\Logs\
Winlogbeat config	C:\Program Files\Winlogbeat\winlogbeat.yml
Winlogbeat log	C:\ProgramData\winlogbeat\Logs\

7.2 Windows Wazuh Service Auto-Recovery

```
# Run PowerShell as Administrator
Set-Service -Name WazuhSvc -StartupType Automatic
Start-Service -Name WazuhSvc

# Restart the service after first, second, and subsequent failures.
sc.exe failure WazuhSvc reset= 86400 actions= restart/60000/restart/60000/restart/60000
sc.exe config WazuhSvc start= auto

# Validate
Get-Service -Name WazuhSvc
Get-Content "C:\Program Files (x86)\ossec-agent\ossec.log" -Tail 80
Test-NetConnection WAZUH_MANAGER_FQDN -Port 1514
Test-NetConnection WAZUH_MANAGER_FQDN -Port 1515
```

7.3 Windows Wazuh Event Channel Collection

```
<localfile>
  <location>Security</location>
  <log_format>eventchannel</log_format>
</localfile>
<localfile>
  <location>System</location>
  <log_format>eventchannel</log_format>
</localfile>
<localfile>
  <location>Application</location>
  <log_format>eventchannel</log_format>
</localfile>
<localfile>
  <location>Microsoft-Windows-PowerShell/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
<localfile>
  <location>Windows PowerShell</location>
  <log_format>eventchannel</log_format>
</localfile>
<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
<localfile>
  <location>Microsoft-Windows-Windows Defender/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
<localfile>
  <location>Microsoft-Windows-TerminalServices-LocalSessionManager/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
<localfile>
  <location>Microsoft-Windows-TerminalServices-RemoteConnectionManager/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
<localfile>
  <location>Microsoft-Windows-TaskScheduler/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
<localfile>
  <location>Microsoft-Windows-WMI-Activity/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
<localfile>
  <location>Microsoft-Windows-WinRM/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
<localfile>
  <location>Microsoft-Windows-AppLocker/EXE and DLL</location>
  <log_format>eventchannel</log_format>
</localfile>
```

```
<localfile>
  <location>Microsoft-Windows-AppLocker/MSI and Script</location>
  <log_format>eventchannel</log_format>
</localfile>
<localfile>
  <location>Microsoft-Windows-Windows Firewall With Advanced Security/Firewall</location>
  <log_format>eventchannel</log_format>
</localfile>
```

7.4 Windows Audit Policy and PowerShell Logging

```
# Run CMD as Administrator
auditpol /set /subcategory:"Logon" /success:enable /failure:enable
auditpol /set /subcategory:"Logoff" /success:enable /failure:enable
auditpol /set /subcategory:"Account Lockout" /success:enable /failure:enable
auditpol /set /subcategory:"User Account Management" /success:enable /failure:enable
auditpol /set /subcategory:"Security Group Management" /success:enable /failure:enable
auditpol /set /subcategory:"Computer Account Management" /success:enable /failure:enable
auditpol /set /subcategory:"Process Creation" /success:enable /failure:enable
auditpol /set /subcategory:"Directory Service Changes" /success:enable /failure:enable
auditpol /set /subcategory:"Audit Policy Change" /success:enable /failure:enable
auditpol /set /subcategory:"Authentication Policy Change" /success:enable /failure:enable
auditpol /set /subcategory:"Sensitive Privilege Use" /success:enable /failure:enable

# Include command line in process creation events
reg add "HKLM\Software\Microsoft\Windows\CurrentVersion\Policies\System\Audit" /v
ProcessCreationIncludeCmdLine_Enabled /t REG_DWORD /d 1 /f
```

```
# PowerShell as Administrator
New-Item -Path "HKLM:\Software\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging" -Force
Set-ItemProperty -Path "HKLM:\Software\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging" -Name
EnableScriptBlockLogging -Value 1

New-Item -Path "HKLM:\Software\Policies\Microsoft\Windows\PowerShell\ModuleLogging\ModuleNames" -Force
Set-ItemProperty -Path "HKLM:\Software\Policies\Microsoft\Windows\PowerShell\ModuleLogging" -Name
EnableModuleLogging -Value 1
Set-ItemProperty -Path "HKLM:
\Software\Policies\Microsoft\Windows\PowerShell\ModuleLogging\ModuleNames" -Name "*" -Value ""
```

7.5 Windows SOC Event ID Matrix

Use Case	Key Event IDs
Successful/failed logon	4624, 4625
Explicit credential use	4648
Admin privilege assigned at logon	4672
User account created/enabled/disabled/deleted	4720, 4722, 4725, 4726
Password reset/change	4724, 4723
Privileged group membership change	4728, 4732, 4756
Account lockout	4740
Kerberos activity / Kerberoasting visibility	4768, 4769
NTLM authentication	4776
Audit log cleared	1102
Service installed	7045
Scheduled task created	4698
Process created	4688
Object access	4663
Directory service object changed	5136
Defender detection/remediation	1116, 1117
PowerShell module/script block	4103, 4104

8. Linux Production Deployment

8.1 Linux Paths

Item	Path / Service
Wazuh config	/var/ossec/etc/ossec.conf
Wazuh log	/var/ossec/logs/ossec.log
Wazuh service	wazuh-agent
Debian/Ubuntu auth/syslog	/var/log/auth.log, /var/log/syslog
RHEL/CentOS/Rocky auth/syslog	/var/log/secure, /var/log/messages
Auditd log	/var/log/audit/audit.log

8.2 Linux Service Auto-Recovery

```
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent

# Add systemd recovery override
sudo mkdir -p /etc/systemd/system/wazuh-agent.service.d
sudo tee /etc/systemd/system/wazuh-agent.service.d/override.conf >/dev/null <<'EOF'
[Service]
Restart=always
RestartSec=10
StartLimitIntervalSec=0
EOF

sudo systemctl daemon-reload
sudo systemctl restart wazuh-agent
systemctl is-enabled wazuh-agent
systemctl status wazuh-agent --no-pager
sudo tail -n 100 /var/ossec/logs/ossec.log
nc -vz WAZUH_MANAGER_FQDN 1514
nc -vz WAZUH_MANAGER_FQDN 1515
```

8.3 Linux Log Collection

```
<!-- Debian/Ubuntu -->
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/syslog</location>
</localfile>
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/auth.log</location>
</localfile>
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/kern.log</location>
</localfile>
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/dpkg.log</location>
</localfile>
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/apt/history.log</location>
</localfile>

<!-- RHEL/CentOS/Rocky/AlmaLinux -->
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/messages</location>
</localfile>
<localfile>
```

```

    <log_format>syslog</log_format>
    <location>/var/log/secure</location>
</localfile>
<localfile>
    <log_format>syslog</log_format>
    <location>/var/log/cron</location>
</localfile>
<localfile>
    <log_format>syslog</log_format>
    <location>/var/log/yum.log</location>
</localfile>
<localfile>
    <log_format>syslog</log_format>
    <location>/var/log/dnf.log</location>
</localfile>

<!-- auditd -->
<localfile>
    <log_format>audit</log_format>
    <location>/var/log/audit/audit.log</location>
</localfile>

<!-- Web and database examples -->
<localfile>
    <log_format>apache</log_format>
    <location>/var/log/apache2/access.log</location>
</localfile>
<localfile>
    <log_format>apache</log_format>
    <location>/var/log/apache2/error.log</location>
</localfile>
<localfile>
    <log_format>syslog</log_format>
    <location>/var/log/nginx/access.log</location>
</localfile>
<localfile>
    <log_format>syslog</log_format>
    <location>/var/log/nginx/error.log</location>
</localfile>
<localfile>
    <log_format>syslog</log_format>
    <location>/var/log/postgresql/postgresql-*.log</location>
</localfile>

```

8.4 Linux auditd SOC Rules

```

# Install auditd
sudo apt install auditd audispd-plugins -y      # Debian/Ubuntu
sudo yum install audit audit-libs -y            # RHEL/CentOS/Rocky
sudo systemctl enable auditd
sudo systemctl start auditd

# /etc/audit/rules.d/soc.rules
-w /etc/passwd -p wa -k identity
-w /etc/shadow -p wa -k identity
-w /etc/group -p wa -k identity
-w /etc/sudoers -p wa -k privilege
-w /etc/sudoers.d/ -p wa -k privilege
-w /etc/ssh/sshd_config -p wa -k ssh_config
-w /root/.ssh/ -p wa -k root_ssh
-w /etc/crontab -p wa -k cron
-w /etc/cron.d/ -p wa -k cron
-w /etc/systemd/system/ -p wa -k systemd_persistence
-w /lib/systemd/system/ -p wa -k systemd_persistence
-w /var/log/ -p wa -k log_tampering
-w /var/log/audit/ -p wa -k audit_log_tampering
-a always,exit -F arch=b64 -S execve -F euid=0 -k root_command
-a always,exit -F arch=b32 -S execve -F euid=0 -k root_command
-a always,exit -F arch=b64 -S init_module,finit_module,delete_module -k kernel_module
-a always,exit -F arch=b32 -S init_module,delete_module -k kernel_module

```



```
sudo augenrules --load
sudo auditctl -l
```

9. macOS Production Deployment

9.1 macOS Paths and Service Commands

Item	Path / Command
Wazuh config	/Library/Ossec/etc/ossec.conf
Wazuh log	/Library/Ossec/logs/ossec.log
Control command	/Library/Ossec/bin/wazuh-control start stop restart status
LaunchDaemon	/Library/LaunchDaemons/com.wazuh.agent.plist
System log	/var/log/system.log
Install log	/var/log/install.log
Wi-Fi log	/var/log/wifi.log
Audit log	/var/audit/current

```
sudo /Library/Ossec/bin/wazuh-control status
sudo /Library/Ossec/bin/wazuh-control start
sudo launchctl load -w /Library/LaunchDaemons/com.wazuh.agent.plist
sudo /Library/Ossec/bin/wazuh-control restart
sudo tail -n 100 /Library/Ossec/logs/ossec.log
nc -vz WAZUH_MANAGER_FQDN 1514
nc -vz WAZUH_MANAGER_FQDN 1515
```

9.2 macOS Log Collection

```
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/system.log</location>
</localfile>
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/install.log</location>
</localfile>
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/wifi.log</location>
</localfile>
<localfile>
  <log_format>syslog</log_format>
  <location>/var/audit/current</location>
</localfile>

<!-- Modern macOS Unified Logging example; tune predicates to avoid excessive EPS. -->
<localfile>
  <location>macos</location>
  <log_format>macos</log_format>
  <query type="log" level="info">process == "sshd" OR process == "sudo" OR eventMessage CONTAINS
"authentication"</query>
</localfile>
```

10. Centralized Agent Configuration

Use Wazuh groups to prevent manual endpoint-by-endpoint edits. Configure group-specific agent.conf files on the manager.

Group	File
Default	/var/ossec/etc/shared/default/agent.conf
Windows	/var/ossec/etc/shared/windows/agent.conf
Linux	/var/ossec/etc/shared/linux/agent.conf
macOS	/var/ossec/etc/shared/macOS/agent.conf
Domain Controllers	/var/ossec/etc/shared/domain-controllers/agent.conf
Database Servers	/var/ossec/etc/shared/database-servers/agent.conf

```
# Example: create a Windows group and assign an agent
sudo /var/ossec/bin/agent_groups -a -g windows
sudo /var/ossec/bin/agent_groups -a -i AGENT_ID -g windows

# Edit centralized config
sudo nano /var/ossec/etc/shared/windows/agent.conf
sudo systemctl restart wazuh-manager
```

Example Windows group agent.conf

```
<agent_config>
  <localfile>
    <location>Security</location>
    <log_format>eventchannel</log_format>
  </localfile>
  <localfile>
    <location>System</location>
    <log_format>eventchannel</log_format>
  </localfile>
  <localfile>
    <location>Microsoft-Windows-PowerShell/Operational</location>
    <log_format>eventchannel</log_format>
  </localfile>
  <localfile>
    <location>Microsoft-Windows-Sysmon/Operational</location>
    <log_format>eventchannel</log_format>
  </localfile>
</agent_config>
```

11. Syslog Forwarding to Wazuh

11.1 Linux rsyslog Forwarding

```
# /etc/rsyslog.d/90-wazuh-forward.conf
# TCP syslog forwarding to Wazuh manager
*. * @@WAZUH_MANAGER_FQDN:514

sudo systemctl restart rsyslog
sudo systemctl enable rsyslog
logger "WAZUH SYSLOG TEST from $(hostname)"
```

11.2 Network Device Syslog Standard

Device Type	Minimum Logs to Forward
-------------	-------------------------

Firewall	traffic allow/deny, NAT, threat/IPS, VPN, admin login, config/policy changes
Switch	admin login, config changes, port up/down, STP, authentication failures
Router	admin login, ACL deny, route changes, interface changes
VPN	successful/failed login, MFA failure, session start/end, source IP, username
Proxy/SWG	URL request, category block, malware block, policy bypass
Email Gateway	phishing, malware, URL rewrite/click, SPF/DKIM/DMARC failure
IDS/IPS/WAF	exploit signatures, scan attempts, C2 attempts, blocked payloads

12. Winlogbeat Optional Production Pipeline

Use Winlogbeat when the architecture requires a separate Windows event pipeline to Logstash/Elastic/another SIEM. For Wazuh-only collection, the Wazuh agent eventchannel configuration is normally sufficient.

C:\Program Files\Winlogbeat\winlogbeat.yml

```
winlogbeat.event_logs:
  - name: Application
    ignore_older: 72h
  - name: System
    ignore_older: 72h
  - name: Security
    ignore_older: 72h
  - name: Windows PowerShell
    ignore_older: 72h
  - name: Microsoft-Windows-PowerShell/Operational
    ignore_older: 72h
    event_id: 400,403,4103,4104,600,800
  - name: Microsoft-Windows-Sysmon/Operational
    ignore_older: 72h
  - name: Microsoft-Windows-Windows Defender/Operational
    ignore_older: 72h
  - name: Microsoft-Windows-TerminalServices-LocalSessionManager/Operational
    ignore_older: 72h
  - name: Microsoft-Windows-TerminalServices-RemoteConnectionManager/Operational
    ignore_older: 72h
  - name: Microsoft-Windows-TaskScheduler/Operational
    ignore_older: 72h
  - name: Microsoft-Windows-WMI-Activity/Operational
    ignore_older: 72h
  - name: Microsoft-Windows-WinRM/Operational
    ignore_older: 72h
  - name: Microsoft-Windows-AppLocker/EXE and DLL
    ignore_older: 72h
  - name: Microsoft-Windows-AppLocker/MSI and Script
    ignore_older: 72h

processors:
  - add_host_metadata:
      when.not.contains.tags: forwarded
  - add_cloud_metadata: ~

output.logstash:
  hosts: ["LOGSTASH_IP:5044"]
```

```
logging.level: info
logging.to_files: true
logging.files:
  path: C:\ProgramData\winlogbeat\Logs
  name: winlogbeat
  keepfiles: 7
```

```
# PowerShell as Administrator
cd "C:\Program Files\Winlogbeat"
.\install-service-winlogbeat.ps1
Set-Service -Name winlogbeat -StartupType Automatic
Start-Service winlogbeat
sc.exe failure winlogbeat reset= 86400 actions= restart/60000/restart/60000/restart/60000
.\winlogbeat.exe test config -c .\winlogbeat.yml -e
.\winlogbeat.exe test output -c .\winlogbeat.yml -e
Get-Service winlogbeat
```

13. File Integrity Monitoring Baseline

13.1 Windows FIM

```
<syscheck>
  <disabled>no</disabled>
  <frequency>43200</frequency>
  <scan_on_start>yes</scan_on_start>
  <directories check_all="yes" realtime="yes">C:\Windows\System32\drivers\etc</directories>
  <directories check_all="yes" realtime="yes">C:\Windows\System32\Tasks</directories>
  <directories check_all="yes" realtime="yes">C:\ProgramData\Microsoft\Windows\Start
Menu\Programs\Startup</directories>
  <windows_registry arch="both">HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run</
windows_registry>
  <windows_registry
arch="both">HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\RunOnce</windows_registry>
  <windows_registry arch="both">HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services</
windows_registry>
</syscheck>
```

13.2 Linux FIM

```
<syscheck>
  <disabled>no</disabled>
  <frequency>43200</frequency>
  <scan_on_start>yes</scan_on_start>
  <directories check_all="yes" realtime="yes">/etc</directories>
  <directories check_all="yes" realtime="yes">/bin</directories>
  <directories check_all="yes" realtime="yes">/sbin</directories>
  <directories check_all="yes" realtime="yes">/usr/bin</directories>
  <directories check_all="yes" realtime="yes">/usr/sbin</directories>
  <directories check_all="yes" realtime="yes">/boot</directories>
  <directories check_all="yes" realtime="yes">/root/.ssh</directories>
</syscheck>
```

13.3 macOS FIM

```
<syscheck>
  <disabled>no</disabled>
  <frequency>43200</frequency>
  <scan_on_start>yes</scan_on_start>
  <directories check_all="yes" realtime="yes">/etc</directories>
  <directories check_all="yes" realtime="yes">/usr/bin</directories>
  <directories check_all="yes" realtime="yes">/usr/sbin</directories>
  <directories check_all="yes" realtime="yes">/Library/LaunchDaemons</directories>
  <directories check_all="yes" realtime="yes">/Library/LaunchAgents</directories>
```

14. Syscollector and SCA Baseline

```
<wodle name="syscollector">
  <disabled>no</disabled>
  <interval>1h</interval>
  <scan_on_start>yes</scan_on_start>
  <hardware>yes</hardware>
  <os>yes</os>
  <network>yes</network>
  <packages>yes</packages>
  <ports all="no">yes</ports>
  <processes>yes</processes>
</wodle>

<sca>
  <enabled>yes</enabled>
  <scan_on_start>yes</scan_on_start>
  <interval>12h</interval>
</sca>
```

15. Required Log Path Matrix

Platform	Required Paths / Channels
Windows	Security, System, Application, Microsoft-Windows-PowerShell/Operational, Windows PowerShell, Microsoft-Windows-Sysmon/Operational, Microsoft-Windows-Windows Defender/Operational, TerminalServices, TaskScheduler, WMI-Activity, WinRM, AppLocker, Windows Firewall
Windows physical EVTX	C:\Windows\System32\winevt\Logs\Security.evtx; System.evtx; Application.evtx; Microsoft-Windows-PowerShell%4Operational.evtx; Microsoft-Windows-Sysmon%4Operational.evtx
Ubuntu/Debian	/var/log/syslog; /var/log/auth.log; /var/log/kern.log; /var/log/dpkg.log; /var/log/apt/history.log; /var/log/audit/audit.log; /var/log/nginx/*.log; /var/log/apache2/*.log
RHEL/Rocky/Alma/CentOS	/var/log/messages; /var/log/secure; /var/log/cron; /var/log/yum.log; /var/log/dnf.log; /var/log/audit/audit.log; /var/log/httpd/*; /var/log/nginx/*.log
macOS	/var/log/system.log; /var/log/install.log; /var/log/wifi.log; /var/audit/current; /Library/Logs; macOS Unified Logging with tuned predicates
Network/Security Devices	Syslog to Wazuh manager TCP 514 preferred; UDP 514 if device cannot support TCP

16. SOC Validation Test Cases

Test	Command / Action	Expected SOC Result
------	------------------	---------------------

Agent connectivity	Restart endpoint or Wazuh agent service	Agent returns Active in dashboard/agent_control
Windows failed login	Attempt invalid login or use test account	4625 alert/log visible
Windows PowerShell logging	Run a benign PowerShell command	4103/4104 logs visible if policy enabled
Windows Sysmon	Start process or run test DNS lookup	Sysmon event 1/22 visible
Linux SSH failure	Attempt invalid SSH login	/var/log/auth.log or /var/log/secure event visible
Linux sudo/root command	Run sudo whoami	auditd/root_command event visible
macOS sudo/auth	Run sudo whoami	system/audit/unified log visible
Syslog	logger "WAZUH SYSLOG TEST" or device test message	Message visible in archives/alerts depending decoder/rule
FIM	Modify monitored test file	FIM event generated
Disconnected agent	Stop agent service on test host beyond threshold	Disconnected event visible

17. Troubleshooting: Agent Disconnected

Symptom	Likely Cause	Fix
Service stopped after reboot	Service startup not automatic	Enable service; configure OS service recovery
Agent running but dashboard disconnected	Network/firewall/DNS path issue	Test TCP 1514; check DNS; check firewall/security group
Authentication/key error	Duplicate or invalid agent key	Remove old agent entry and re-enroll with unique name
Intermittent disconnects	UDP loss, VPN flap, endpoint sleep	Use TCP 1514; disable sleep on servers; tune reconnect
Pending state persists	Enrollment or first communication incomplete	Check 1515/1514 reachability and manager ossec.log
No Windows logs	Wrong channel name or audit policy disabled	Validate with Get-WinEvent; enable audit policy
No Linux logs	Path does not exist on distro	Use distro-specific paths: auth.log vs secure, syslog vs messages
High EPS/CPU	Too many verbose channels/predicates	Filter noisy channels, tune Sysmon and macOS ULS predicates
Syslog not received	Listener not enabled/source not allowed	Check remote syslog config and allowed-ips

18. Broken Agent Re-Enrollment

18.1 Manager Side

```
sudo /var/ossec/bin/manage_agents -l
sudo /var/ossec/bin/manage_agents -r AGENT_ID
sudo systemctl restart wazuh-manager
```

18.2 Linux Agent

```
sudo systemctl stop wazuh-agent
```

```
sudo /var/ossec/bin/agent-auth -m WAZUH_MANAGER_FQDN -A UNIQUE_AGENT_NAME
sudo systemctl restart wazuh-agent
```

18.3 Windows Agent

```
Stop-Service WazuhSvc
& "C:\Program Files (x86)\ossec-agent\agent-auth.exe" -m WAZUH_MANAGER_FQDN -A UNIQUE_AGENT_NAME
Restart-Service WazuhSvc
```

18.4 macOS Agent

```
sudo /Library/Ossec/bin/wazuh-control stop
sudo /Library/Ossec/bin/agent-auth -m WAZUH_MANAGER_FQDN -A UNIQUE_AGENT_NAME
sudo /Library/Ossec/bin/wazuh-control start
```

19. Daily SOC Health Operations

Frequency	Activity	Owner	Evidence
Daily	Review disconnected agents by criticality	SOC L1/L2	Dashboard screenshot/export
Daily	Check new endpoint enrollment failures	SIEM Engineer	Manager ossec.log/API query
Daily	Review top noisy log sources and EPS spikes	SIEM Engineer	Dashboard/report
Weekly	Validate Sysmon/PowerShell/auditd test coverage on sample endpoints	SOC Engineering	Test case output
Weekly	Review agents not seen > 7 days and decommission stale entries	SOC Manager / IT Ops	Agent inventory list
Monthly	Review group agent.conf baseline and update log onboarding matrix	SIEM Engineer	Change record
Quarterly	Run tabletop test: manager restart, endpoint reboot, network outage	SOC + Infra	Test evidence and lessons learned

20. Production Acceptance Checklist

- ☐ Wazuh manager listens on TCP 1514 and enrollment TCP 1515.
- ☐ Syslog listener TCP/UDP 514 is enabled only if required and restricted by allowed source ranges.
- ☐ All agents use TCP 1514 and valid manager FQDN/IP.
- ☐ Agent notify_time is lower than manager agents_disconnection_time.
- ☐ Windows/Linux/macOS services are set to auto-start and auto-recover.
- ☐ Each endpoint has a unique agent identity and correct group assignment.

- ☐ Windows Security/System/Application/PowerShell/Sysmon/Defender/RDP/WMI/WinRM logs are collected.
- ☐ Linux auth/syslog/secure/messages/auditd/cron/package/web/database logs are collected where applicable.
- ☐ macOS system/install/wifi/audit/unified logging collection is configured with tuned predicates.
- ☐ FIM, SCA, and syscollector are enabled with production-safe intervals.
- ☐ Winlogbeat is deployed only where the target architecture requires it.
- ☐ SOC validation test cases have been executed and evidence saved.
- ☐ Disconnected agent alerting and daily health review process are operational.

21. Official Reference Sources

- Wazuh agent life cycle and disconnected state:
<https://documentation.wazuh.com/current/user-manual/agent/agent-enrollment/agent-life-cycle.html>
- Wazuh client configuration: notify_time, time-reconnect, auto_restart:
<https://documentation.wazuh.com/current/user-manual/reference/ossec-conf/client.html>
- Wazuh global configuration and agents_disconnection_time: <https://documentation.wazuh.com/current/user-manual/reference/ossec-conf/global.html>
- Wazuh localfile configuration for files, Windows event channels, macOS ULS, and journald:
<https://documentation.wazuh.com/current/user-manual/reference/ossec-conf/localfile.html>
- Wazuh log data collection configuration and centralized agent.conf:
<https://documentation.wazuh.com/current/user-manual/capabilities/log-data-collection/monitoring-log-files.html>
- Wazuh OS log collection examples: <https://documentation.wazuh.com/current/user-manual/capabilities/log-data-collection/configuration.html>
- Elastic Winlogbeat event log configuration: <https://www.elastic.co/docs/reference/beats/winlogbeat/configuration-winlogbeat-options>
- Elastic Winlogbeat installation/configuration: <https://www.elastic.co/docs/reference/beats/winlogbeat/winlogbeat-installation-configuration>